

ROLL NO:250701573

EXPERIMENT NO: 6 EXPERIMENT TO UNDERSTAND THE FEATURE SCALING

PROGRAM:

```
import numpy as np
```

```
import pandas as pd
```

```
df = pd.DataFrame({  
    "Country": ["India", "Japan", "China", "India", "Japan",  
               "China", "India", "China", "Japan", "India"],  
    "Age": [22, 31, 28, 35, 42, 26, 39, 45, 30, 34],  
    "Salary": [35000, 52000, 46000, 61000, 75000,  
              41000, 68000, 82000, 50000, 57000]  
})
```

```
print("Original Dataset:")
```

```
print(df)
```

```
country = pd.get_dummies(df["Country"], dtype=int)
```

```
data = pd.concat([country, df[["Age", "Salary"]]], axis=1)
```

```
print("\nPrepared Dataset:")
```

```
print(data)
```

```
mean = data.mean()
```

```
std = data.std()
```

```
standard_scaled = (data - mean) / std
```

```
print("\nStandard Scaled Data:")
```

```
print(standard_scaled)
```

```
minimum = data.min()
```

```
maximum = data.max()
```

```
minmax_scaled = (data - minimum) / (maximum - minimum)
```

```
print("\nMin-Max Scaled Data:")
```

```
print(minmax_scaled)
```

OUTPUT:

Original Dataset:

	Country	Age	Salary
0	India	22	35000
1	Japan	31	52000
2	China	28	46000
3	India	35	61000
4	Japan	42	75000
5	China	26	41000
6	India	39	68000
7	China	45	82000
8	Japan	30	50000
9	India	34	57000

Prepared Dataset:

	China	India	Japan	Age	Salary
0	0	1	0	22	35000
1	0	0	1	31	52000
2	1	0	0	28	46000
3	0	1	0	35	61000
4	0	0	1	42	75000
5	1	0	0	26	41000
6	0	1	0	39	68000

7	1	0	0	45	82000
8	0	0	1	30	50000
9	0	1	0	34	57000

Standard Scaled Data:

	China	India	Japan	Age	Salary
0	-0.621059	1.161895	-0.621059	-1.543951	-1.448420
1	-0.621059	-0.774597	1.449138	-0.303276	-0.313713
2	1.449138	-0.774597	-0.621059	-0.716834	-0.714198
3	-0.621059	1.161895	-0.621059	0.248135	0.287014
4	-0.621059	-0.774597	1.449138	1.213104	1.221479
5	1.449138	-0.774597	-0.621059	-0.992540	-1.047935
6	-0.621059	1.161895	-0.621059	0.799546	0.754246
7	1.449138	-0.774597	-0.621059	1.626662	1.688711
8	-0.621059	-0.774597	1.449138	-0.441129	-0.447208
9	-0.621059	1.161895	-0.621059	0.110282	0.020024

Min-Max Scaled Data:

	China	India	Japan	Age	Salary
0	0.0	1.0	0.0	0.000000	0.000000
1	0.0	0.0	1.0	0.391304	0.361702
2	1.0	0.0	0.0	0.260870	0.234043
3	0.0	1.0	0.0	0.565217	0.553191
4	0.0	0.0	1.0	0.869565	0.851064
5	1.0	0.0	0.0	0.173913	0.127660
6	0.0	1.0	0.0	0.739130	0.702128
7	1.0	0.0	0.0	1.000000	1.000000
8	0.0	0.0	1.0	0.347826	0.319149
9	0.0	1.0	0.0	0.521739	0.468085

